

Stage 15: Returns, Refunds, RMAs & Reverse Logistics

1. Overview & Architecture

Stage 15 establishes an automated Return Merchandise Authorization (RMA) and reverse logistics pipeline supporting category/vendor return policy governance, multi-item return initiation with photo proof, vendor approval/rejection workflows, reverse waybill (AWB) carrier booking, warehouse quality check inspection, automated inventory restocking, and instant escrow refund adjustments.

MERMAID

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flowchart TD
    Buyer[Buyer / Customer] -->|Initiate Return| RMA[RmaService: createRmaRequest]
    RMA --> Policy{Check RMA Policy\nWindow / Category / Restocking Fee}
    Policy -->|Allowed| MasterRMA[rma_requests & rma_items]
    MasterRMA --> Evt1[RmaEvent: REQUESTED]

    subgraph Vendor Workflow
        MasterRMA -->|Review| VReview[Vendor Review: APPROVED / REJECTED]
        VReview -->|Schedule Pickup| RevLog[Reverse Carrier Booking\nGenerate REV-AWB]
    end

    subgraph Quality Inspection & Resolution
        RevLog --> QC[Warehouse Inspection]
        QC --> Restock[InventoryService: ADJUSTMENT_ADD]
        QC --> Refund[SettlementService: refundEscrow]
        Refund --> Complete[RmaStatus: REFUND_PROCESSED]
    end

    end
```

2. Key Components & Implementation Details

A. Database Schema Migration (`V41__returns_and_rma_enhancements.sql`)

- Added Pickup Address and Rejection Details to `rma_requests` :
 - `customer_phone VARCHAR(30)` - `pickup_address_line1 VARCHAR(255)` - `pickup_city VARCHAR(100)` - `pickup_pincode VARCHAR(20)` - `rejection_reason VARCHAR(255)`
- Performance Indexes:
 - `idx_rma_requests_vendor_status` on `rma_requests(vendor_id, status)` -
 - `idx_rma_requests_user_created` on `rma_requests(user_id, created_at DESC)` -
 - `idx_rma_requests_order_vendor` on `rma_requests(order_id, vendor_order_id)` -
 - `idx_rma_events_rma_created` on `rma_events(rma_id, created_at DESC)`

B. RMA Policy Governance (`RmaPolicyServiceImpl.java`)

- Evaluates category and vendor specific policies (`return_window_days` , `is_returnable` , `restocking_fee_percentage` , `allow_refund` , `allow_replacement`).

- Enforces return window checks relative to order delivery timestamp.

C. Master Return Request Lifecycle (`RmaServiceImpl.java`)

- Return Initiation:**
 - Validates purchase history, quantity limits, and item returnability. - Generates unique return identifier (`RMA-YYYY-XXXXXX`). - Calculates line-item refunds and tracks proof images.
- Vendor Review:**
 - Supports vendor approval or rejection with detailed review notes.
- Reverse Logistics Scheduling:**
 - Generates reverse tracking numbers (`REV-CARRIER-XXXXXXIN`) and schedules pickup dates.

D. Warehouse Quality Inspection & Automated Restocking

- QC Assessment:**
 - Assesses returned item condition (`UNOPENED` , `OPENED_UNUSED` , `DAMAGED` , `DEFECTIVE`). - If restock action is `RESTOCK_AVAILABLE` , automatically executes `InventoryService.adjustStock(ADJUSTMENT_ADD)` to replenish warehouse stock.
- Refund & Settlement Execution:**
 - Automatically invokes `SettlementService.refundEscrow` to debit vendor escrow and credit the refund amount. - Transitions RMA status to `REFUND_PROCESSED` and captures `completedAt` timestamp.

3. Verification & Test Suite

- Comprehensive end-to-end integration test suite implemented in `ReturnsAndRmaIntegrationTest.java` :
 - `testRmaPolicyEvaluation` : Policy retrieval, vendor overrides, return window validation.
 - `testCustomerInitiateRmaRequest` : Customer return request initiation, line item validation, and audit event creation.
 - `testVendorRmaLifecycleAndQualityInspection` : Vendor approval, reverse pickup AWB generation, warehouse QC inspection, inventory replenishment, and escrow refund deduction.
 - `testRmaCancellationAndSecurity` : Customer cancellation flow and tenant isolation security.
- Build compilation: `BUILD SUCCESS` across 59 test classes.